CHAPTER 3:

Big Crunch

> Let’s start with the end of the world, why don’t we? Get it over with and move on to more interesting things.
>
> N. K. Jemisin, *The Fifth Season*

On a dark, moonless night in autumn, in the Northern Hemisphere, look up and find the wide W shape of the constellation Cassiopeia. Stare at the space below it for a few seconds and, if the sky is dark enough, you’ll see a faint fuzzy blur almost as wide across as the full Moon. That blur is the Andromeda Galaxy, a great spiral disk of about a trillion stars and a supermassive black hole, all of which are hurtling toward us at 110 kilometers per second.

In about four billion years, Andromeda and our own Milky Way galaxy will collide, creating a brilliant light show. Stars will be flung chaotically out of their orbits, forming stellar streams that stretch across the cosmos in graceful arcs. The sudden smashing together of galactic hydrogen will spark a minor explosion of star birth. Gas will ignite around the previously dormant central supermassive black holes, which will meet in the middle of everything and spiral into each other. Jets of intense radiation and high-energy particles will pierce the chaotic tangle of gas and stars, while the central region of the new Milkdromeda galaxy is irradiated with the X-ray-hot glow of a whirlpool of doomed matter falling into the new, even more supermassive black hole.

Even in the midst of this great galactic train wreck, the vast distances between stars will make head-on stellar impacts unlikely: the Solar System as a whole will probably survive, more or less. Not the Earth, though. By that time, the Sun will have already begun to swell to red-giant size, heating up the Earth enough to boil the oceans and completely sterilize the surface of all possibility of life. If, however, any outpost of human ingenuity manages to sustain a presence in the Solar System to watch, the combining of the two great spiral galaxies will be an awe-inspiring and beautiful process, playing out over billions of years. When the particle jets and